

SATYAM CHAURASIA

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Education

Arizona State University

Aug. 2024 – May 2026

Master of Science in Software Engineering

Tempe, Arizona

courses: Advanced Data Structures, Statistical Machine Learning, Software Project Management, Data Processing/Mining

Uttarakhand Technical University

Oct. 2020 – May 2024

B.Tech in Computer Science

Dehradun, Uttarakhand

courses: Data Structures, database management systems, Operating Systems, Computer Networks, Software Testing

Experience

Software Development Intern

Jun 2025 – Aug 2025

Shoptaki

Remote, New York US

- Researched and reverse-engineered the LangChain framework, mapping core abstractions (chains, tools, agents, memory) to guide design of a SmartChain-native alternative.
- Implemented multiple tool-using AI agents (GPT-4 + tools) for document Q&A, compliance support, and workflow automation; evaluated agentic patterns for reliable orchestration.

Software Development Intern

Jun 2023 – Aug 2023

Escalera Technologies

Lucknow, UP India

- Contributed to a containerized real-time data pipeline using Python, Kafka, Kafka Connect, Kubernetes, Docker, and Neo4j to stream taxi-trip events into Neo4j for graph storage.
- Configured Kafka Connect sink workflows and Kubernetes services, troubleshooting Kafka listeners, service exposure, pod restarts, and resource limits while supporting graph analytics workflows using PageRank and BFS.

Projects

Affinity Proto – AI Social Discovery Platform | [GitHub](#)

May 2026

Next.js, React, TypeScript, FastAPI, SQLAlchemy, PostgreSQL, pgvector, sentence-transformers

- Built a full-stack AI-powered social discovery prototype that matches users through semantic similarity of published thoughts using Next.js, React, FastAPI, PostgreSQL, and pgvector.
- Implemented JWT authentication, bcrypt password hashing, protected REST APIs, ownership-based authorization, and thought CRUD with draft/public visibility controls.
- Designed a semantic matching pipeline that converts public thoughts into user-level vector embeddings and ranks nearby users using pgvector cosine similarity.

K-12 Educational Image Search Chatbot | [GitHub](#)

April 2026

Python, Ollama, OpenCLIP, ChromaDB, Wikimedia Commons API, Vector Databases, LLMs, NLP

- Developed a multimodal image retrieval extension for a K-12 chatbot that extracts intent, context, and search keywords from user prompts to fetch relevant educational images.
- Integrated internet-based image retrieval into the chatbot pipeline, displaying contextually relevant images alongside generated LLM responses.
- Added Ollama-based local LLM support for offline execution and contributed to the OpenCLIP and ChromaDB semantic image search workflow.

Reddit Data Ingestion Pipeline | *PostgreSQL, SQL, Bash, pg_bulkload*

February 2025

- Designed a relational schema for the curated Pushshift Reddit dataset and implemented **primary/foreign keys + constraints** to enforce integrity.
- Automated end-to-end setup with a **Bash runner** invoking modular SQL scripts to reliably reproduce DB creation, loading, and query outputs.
- Optimized ingestion of **10,557,466 comments** using **pg_bulkload** and generated persisted analytics outputs by materializing results into **query1–query5 tables**.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, Bash, C

Frontend: React, Next.js, Tailwind CSS

Backend & APIs: FastAPI, REST APIs, SQLAlchemy, Alembic, Pydantic, JWT

Databases & AI/ML: PostgreSQL, MySQL, Neo4j, pgvector, ChromaDB, LangChain, sentence-transformers, Ollama, OpenCLIP, OpenAI API

DevOps & Tools: Docker, Kubernetes, Helm, Apache Kafka, Kafka Connect, Git, JUnit 5, JaCoCo, Jira

Achievements

- Won **1st Prize** at **TECHNOTHON '22** (Hackathon) for developing a prototype wheelchair for mobility assistance.